CausalVariant#

https://pytorch.org/docs/stable/generated/torch.nn.attention.bias.CausalVari

Description:

Enum for causal variants used in attention mechanisms. Defines two types of causal biases: UPPER_LEFT: Represents upper-left triangular bias for standard causal attention. The equivalent pytorch code for constructing this bias is: For instance, with shape=(3,4), the materialized bias tensor will be: LOWER_RIGHT: Represents lower-right triangular bias, the include values are aligned to the lower right corner of the matrix.

Function Signature

```
class torch.nn.attention.bias.CausalVariant(value)[source]#
```

Code Examples

```
torch.tril(torch.ones(size, dtype=torch.bool))
```

```
[[1, 0, 0, 0],
 [1, 1, 0, 0],
 [1, 1, 1, 0]]
```

```
diagonal_offset = size[1] - size[0]
torch.tril(
    torch.ones(size, dtype=torch.bool),
    diagonal=diagonal_offset,
)
```

```
[[1, 1, 0, 0],
 [1, 1, 1, 0],
 [1, 1, 1, 1]]
```

Notes & Warnings

⚠ **WARNING** Warning This enum is a prototype and subject to change.

Detailed Documentation

Documentation

Enum for causal variants used in attention mechanisms.

Defines two types of causal biases:

UPPER_LEFT: Represents upper-left triangular bias for standard causal attention. The equivalent pytorch code for constructing this bias is:

For instance, with shape=(3,4), the materialized bias tensor will be:

LOWER_RIGHT: Represents lower-right triangular bias, the include values are aligned to the lower right corner of the matrix.

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For instance, with shape=(3,4), the materialized bias tensor will be:

Note that these variants are equivalent to each other when the sequence lengths of the query and key/value tensors are equal since the triangular matrix is square.

This enum is a prototype and subject to change.